

Objectives

- ◆ In this session, you will learn to:
 - ◆ Introduce Model View Controller (MVC)

Introducing MVC

- ◆ While designing a Web application, you need to ensure that the Web application meets the following goals:
 - ◆ Extensibility and maintainability
 - ◆ Division of work
 - ◆ Scalability
 - ◆ Code reusability
- ◆ For this, you can develop a Web application on the basis of a design pattern.

Introducing MVC (Contd.)

- ◆ A design pattern can be defined as a set of proven solutions for common programming problems.
- ◆ Some of the benefits of using design patterns are:
 - ◆ Proven design
 - ◆ Reusable
 - ◆ Expressive
- ◆ MVC is one such design pattern, which is used widely in Web application development.
- ◆ The MVC design pattern enables the development of Java EE applications with segregation of the data access, presentation, and business logic of an application into the following components:
 - ◆ Model
 - ◆ View
 - ◆ Controller

The Model

- ◆ The model:
 - ◆ represents information about a website or Web application.
 - ◆ is a non-visual object and contains all the data that is used for the user interface.
 - ◆ is created using JavaBeans.

The View

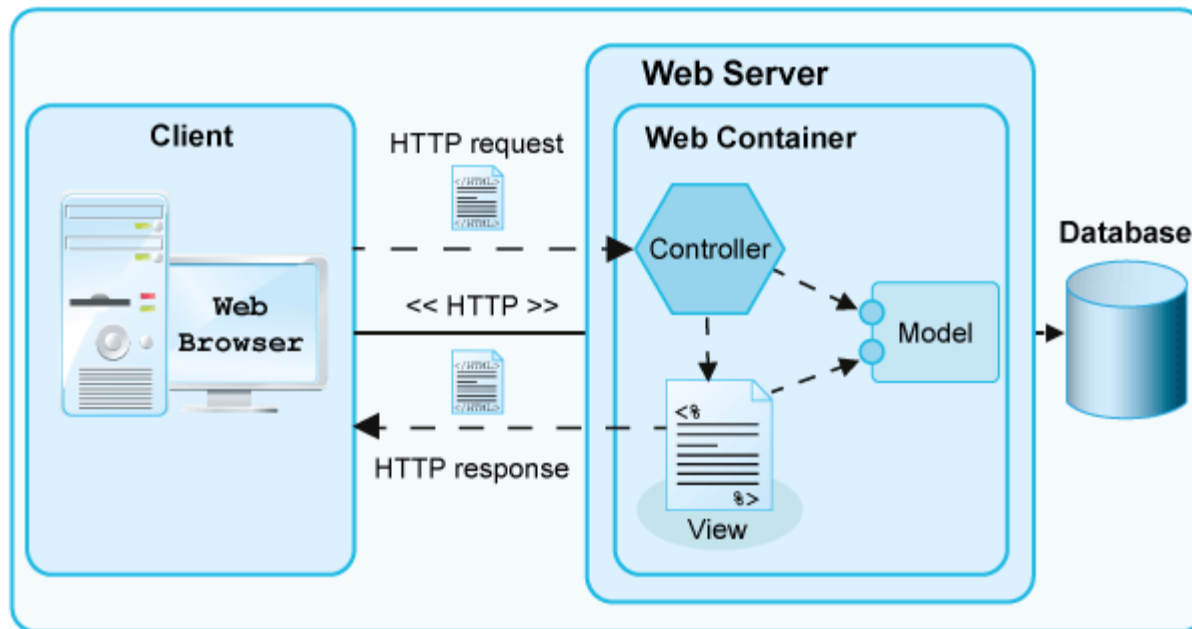
- ◆ The view:
 - ◆ represents the display of the information provided by the model in a UI.
 - ◆ can be used to display the information provided by the model in different forms.

The Controller

- ◆ The controller:
 - ◆ handles the requests received from the view, and the data received from the model.
 - ◆ controls the entire data flow of an application.
 - ◆ is primarily created using servlets.

The Controller (Contd.)

- ◆ The following figure shows the interaction between a model, view, and a controller in an application.



Just a minute

- ◆ The Java Web applications developed using the MVC design pattern use _____ to create the controller.

- ◆ Solution:
 - ◆ servlets

Summary

- ◆ In this session, you learned that:
 - ◆ While designing a Web application, you need to ensure that the Web application meets the following goals:
 - ◆ Extensibility and maintainability
 - ◆ Division of work
 - ◆ Scalability
 - ◆ Code reusability
 - ◆ To save time and effort, you should record some proven solutions to common problem areas. Such common solutions are called design patterns.
 - ◆ Design patterns are well-thought out and previously-used solutions to programming problems.
 - ◆ The most commonly used design pattern by the Web developers today is MVC.

Summary (Contd.)

- ◆ The MVC design pattern enables the development of Java EE applications with segregation of the data access, presentation, and business logic into Model, View, and Controller, respectively.
- ◆ The Java Web applications developed using the MVC design pattern uses JavaBeans or Java classes to create the model.
- ◆ The Java Web applications developed using the MVC design pattern uses JSP to create the view.
- ◆ The Java Web applications developed using the MVC design pattern uses servlets to create the controller.